



EE 603: Renewable Energy and Smart Grid

Course Code	EE 603
Course Name	Renewable Energy and Smart Grid
L-T-P-C	3-0-0-3
Prerequisites	EE 203 Electromechanics, EE 303 Power Systems
Intended for	UG/MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

Basic concepts, definitions and classifications of energy resources; grid code and characteristics; electrical output characteristics of various renewable energy sources; compatibility issues and options.

Introduction to major RES, grid integration issues, challenges and methodologies, power electronics converters for grid integration, hybrid systems and virtual power plants, storage, Cost of interconnection and responsibilities, forecasting, scheduling of RES, regulatory issues and energy markets. Introduction to smart grid concept.

Text and Reference Books

1. Ali Kehani, Mohammad N. Marwali, Min Dai, **Integration of Green and Renewable Energy in Electric Power Systems**, Wiley Interscience, 2009.
2. Gil Masters, **Renewable and Efficient Electric Power Systems**, Wiley-IEEE Press, 2004.
3. Felix A. Farret, M. Godoy Simoes, **Integration of Alternative Sources of Energy**, IEEE Press and Wiley Interscience Publication, 2006.
4. John Twidell and Tony Weir, **Renewable Energy Resources**, Talyor and Francis, 2006.
5. T. Ackermann, **Wind Power in Power Systems**, John Wiley
6. Other Sources: IEEE Transactions on Smart Grid